

scheduling and assigning tasks to tracking performance, finance and team tuning, making it a one-stop solution for all organisational needs.

Recently, small businesses have started adopting SaaS based online cloud project management solutions as these keep track of all the key variables and help companies to deliver their projects on time and within budget. With the implementation of high-end security, cloud hosting and unlimited integration, various project management solutions are becoming more affordable, and many of them are completely free to use.

Let's now explore effective open source project management tools that businesses can implement. These tools are open source and currently deployed in various startups, SMEs and even in some large scale organisations. By researching the extent of their implementation in the real world, we arrived at the following top five open source project management tools that provide a comprehensive set of useful features for small businesses.

OpenProject

This is a Web based project management solution which was designed using Ruby on Rails and Angular JS. It was released under the GNU General Public License Version 3 and is under consistent development by the open source community. It is highly suitable for location-independent team collaboration.

OpenProject is ideal for project teams to work in throughout the project's life cycle. The platform offers various features like collaborative project planning, timeline reports, management of tasks, reporting of project costs incurred from time to time, Scrum and much more.

This open source project management tool is popular because of its intuitive user interface, powerful documentation and enhanced features.

Features

- **Project planning and scheduling:** OpenProject lets project managers easily define the project's objectives and work specifications. The tool analyses the required activities and creates a professional project plan that demonstrates how and when the project will be delivered, on time and within the scope.
- **Product roadmap and release planning:** This feature enables project managers to visualise, do proper planning and communicate the project's activities to the team members. It also helps to strategise the project roadmap by enabling transparency among these members.
- **Effective task management and team collaboration:** OpenProject is a powerful tool to enable teams to keep consistent track of the work done as well as the work pending, and get effective results. It enables the team members to organise and assign tasks, and effectively communicate these at any single point of time.

- **Agile and Scrum integration:** Considering that agile based projects have a shorter development period, OpenProject provides a Scrum based platform for agile teams to build, measure and learn on. The teams can prioritise and do effective task tracking. This feature integrates high-end modules like roadmap planning, bug tracking and task management.
 - **Time tracking, budgeting and reporting of costs:** OpenProject creates an ideal platform to build projects and effectively track all activities. Progress can be tracked from a cost perspective to enable timely budgeting.
 - **Bug tracking:** OpenProject provides add-on benefits by ensuring quality via tracking tasks and through effective communication. With OpenProject, quality assurance managers can easily capture, differentiate and prioritise the bugs in projects.
- In addition to the above, OpenProject provides add-on features via cloud linkage especially designed for large scale enterprises.

Latest version: 7.2

Official website: <https://www.openproject.org/>

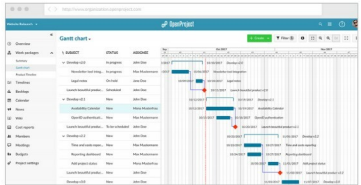


Figure 1: The OpenProject GUI

ProjectLibre

ProjectLibre is open source, free to use project management software developed by Marc O'Brien as an alternative to Microsoft Project. It's used for task management, resource allocation, tracking of tasks, Gantt charts and much more.

ProjectLibre is developed using Java and provides a similar user interface as Microsoft Project. It includes a ribbon-style menu and the same series of steps to create a project plan, i.e., create an indented task list or 'work breakdown structure' (WBS), set durations, create links, assign resources, etc.

ProjectLibre can easily run on all platforms like Windows, Linux and Mac OS.

Features

- **Compatible with MS Project:** ProjectLibre is fully compatible with MS Project 2003 (till date version files). It has almost the same user interface and approach for creating a project plan. It enables project managers to do the following: create an indented task list, work breakdown structure, project